q78i3gzfi

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0.1 ADT Lab2

Shivali Mate Link: https://colab.research.google.com/drive/1m_2mKmKDN-dMWQ_xai-Q5_xY7lqoObYE

[ ]:
```
#importing required libraries
import sqlite3
import pandas as pd
from IPython.display import display
```

[ ]:
```
# Create a connection to the SQLite database (1 points)
conn = sqlite3.connect('NetFlix.db')
c = conn.cursor()
```

[ ]:
```
# Create a connection to the SQLite database (1 points)
conn2 = sqlite3.connect('spotify_top50_2021.db')
c = conn.cursor()
```

[ ]:
```
# Load data from CSV files (1 points)
spotify_df = pd.read_csv("spotify_top50_2021.csv")
netflix_df = pd.read_csv('NetFlix.csv')
```

[ ]:
```
# Create tables and insert data
spotify_df.to_sql('spotify_songs', conn, if_exists='replace', index=False)
netflix_df.to_sql('netflix_shows', conn, if_exists='replace', index=False)
```

[ ]:
```
7787
```

0.2 Q1. Given a Spotify dataset containing various metrics for songs, retrieve the top 5 songs ranked by their popularity. (1 points)

[ ]:
```
c.execute('''select track_name, artist_name, popularity
from spotify_songs
order by popularity desc
limit 5;

''')
```

```
df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Popularity'])
display(df)
```

	Track	Artist	Popularity
0	good 4 u	Olivia Rodrigo	95
1	Bad Habits	Ed Sheeran	95
2	Woman	Doja Cat	95
3	Heat Waves	Glass Animals	94
4	Yonaguni	Bad Bunny	94

##Q2. Identify and list all songs from the dataset that have a danceability score exceeding 0.7. Ensure that each song included meets this criterion based on its danceability rating. (1 points)

[ ]:
```
c.execute('''select track_name, artist_name, danceability
from spotify_songs
where danceability > 0.7;

''')
df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Danceability'])
display(df)
```

	Track	Artist	\
0	Levitating (feat. DaBaby)	Dua Lipa	
1	Kiss Me More (feat. SZA)	Doja Cat	
2	Heat Waves	Glass Animals	
3	Beggin'	Måneskin	
4	Astronaut In The Ocean	Masked Wolf	
5	DÁKITI	Bad Bunny	
6	INDUSTRY BABY (feat. Jack Harlow)	Lil Nas X	
7	Bad Habits	Ed Sheeran	
8	Butter	BTS	
9	Todo De Ti	Rauw Alejandro	
10	Mood (feat. iann dior)	24kGoldn	
11	The Business	Tiësto	
12	Dynamite	BTS	
13	Friday (feat. Mufasa & Hypeman) - Dopamine Re-…	Riton	
14	Pepas	Farruko	
15	positions	Ariana Grande	
16	Bandido	Myke Towers	
17	I WANNA BE YOUR SLAVE	Måneskin	
18	RAPSTAR	Polo G	
19	LA NOCHE DE ANOCHE	Bad Bunny	
20	Streets	Doja Cat	
21	Fiel	Los Legendarios	
22	Don't Start Now	Dua Lipa	
23	Lemonade (feat. Gunna Don Toliver & NAV)	Internet Money	
24	Woman	Doja Cat	
25	Qué Más Pues?	J Balvin	

```
26                          Head & Heart (feat. MNEK)       Joel Corry
27                                              34+35    Ariana Grande
28                                   Pareja Del Año  Sebastian Yatra

    Danceability
0          0.702
1          0.764
2          0.761
3          0.714
4          0.778
5          0.731
6          0.741
7          0.807
8          0.759
9          0.780
10         0.701
11         0.798
12         0.746
13         0.824
14         0.762
15         0.737
16         0.713
17         0.750
18         0.789
19         0.856
20         0.749
21         0.849
22         0.793
23         0.800
24         0.824
25         0.903
26         0.734
27         0.830
28         0.715
```

##Q3. Given a dataset from Spotify that includes various artists and their popularity scores, determine the average popularity score for each artist. This will help understand how popular each artist is on average across the dataset. (1 points)

[ ]:
```
c.execute('''select artist_name, avg(popularity)
from spotify_songs
group by artist_name;

''')
df = pd.DataFrame(c.fetchall(), columns=['Artist', 'Avg Popularity'])
display(df)
```

```
                Artist  Avg Popularity
```

0	24kGoldn	89.000000
1	Ariana Grande	86.500000
2	BTS	69.000000
3	Bad Bunny	89.333333
4	Bruno Mars	86.000000
5	Doja Cat	90.750000
6	Dua Lipa	86.500000
7	Duncan Laurence	78.000000
8	Ed Sheeran	95.000000
9	Farruko	86.000000
10	Giveon	86.000000
11	Glass Animals	94.000000
12	Harry Styles	90.000000
13	Internet Money	81.000000
14	J Balvin	79.000000
15	Joel Corry	86.000000
16	Justin Bieber	90.000000
17	Kali Uchis	87.000000
18	Lewis Capaldi	88.000000
19	Lil Nas X	90.000000
20	Los Legendarios	82.000000
21	Masked Wolf	77.000000
22	Myke Towers	82.000000
23	Måneskin	90.500000
24	Olivia Rodrigo	92.250000
25	Polo G	82.000000
26	Rauw Alejandro	69.000000
27	Riton	88.000000
28	SZA	86.000000
29	Sebastian Yatra	86.000000
30	Tate McRae	88.000000
31	The Kid LAROI	88.500000
32	The Neighbourhood	92.000000
33	The Weeknd	91.500000
34	Tiësto	88.000000

##Q4. Identify artists who have more than one song in the top 50 charts and count how many songs each of these artists have in the top 50. (2 points)

[ ]:
```
c.execute('''select artist_name, count(*) as song_count
from spotify_songs
group by artist_name having song_count > 1
;

''')
df = pd.DataFrame(c.fetchall(), columns=['Artist', 'Song_Count'])
display(df)
```

```
           Artist  Song_Count
0   Ariana Grande           2
1             BTS           2
2       Bad Bunny           3
3        Doja Cat           4
4        Dua Lipa           2
5       Lil Nas X           2
6        Måneskin           2
7  Olivia Rodrigo           4
8   The Kid LAROI           2
9      The Weeknd           2
```

##Q5. Identify and list all the unique genres present in the Netflix dataset to understand the variety of content offered. (1 points)

[ ]:
```
c.execute(''' select distinct genres
from netflix_shows

''')
df = pd.DataFrame(c.fetchall(), columns=['Genre'])
display(df)
```

```
                                                Genre
0    International TV Shows, TV Dramas, TV Sci-Fi &…
1       Horror Movies, International Movies, Thrillers
2          Dramas, International Movies, Sports Movies
3                                             Comedies
4    British TV Shows, Docuseries, Science & Nature TV
..                                                 …
487          Kids' TV, Reality TV, Science & Nature TV
488  International TV Shows, Spanish-Language TV Sh…
489                                  Sci-Fi & Fantasy
490  International TV Shows, TV Action & Adventure,…
491  Stand-Up Comedy & Talk Shows, TV Mysteries, TV…

[492 rows x 1 columns]
```

##Q6. Find all Netflix shows with “Blue” in the title. (1 points)

[ ]:
```
c.execute(''' select title, genres
from netflix_shows
where title LIKE '%Blue%';

''')
df = pd.DataFrame(c.fetchall(), columns=['Title', 'Genre'])
display(df)
```

```
                                           Title  \
0          Blue Mountain State: The Rise of Thadland
```

```
1                                      Blue Planet II
2                                           Blue Ruin
3                                         Blue Streak
4                                      Blue Valentine
5   Brave Blue World: Racing to Solve Our Water Cr…
6                  Colin Quinn: Red State Blue State
7                     GRANBLUE FANTASY the Animation
8              Hunter in the Blue Side of Manchester
9                                        Mission Blue
10                                       Red vs. Blue
11                                The Blue Elephant 2
12   The Blue Planet: A Natural History of the Oceans
13                                  The Blue Umbrella
14                 Auschwitz: Blueprints of Genocide
15                                      Blue Exorcist
16                         Blue Is the Warmest Color
17                                       Blue Jasmine
18                                           Blue Jay

                                                 Genre
0                                               Comedies
1   British TV Shows, Docuseries, Science & Nature TV
2                       Independent Movies, Thrillers
3                        Action & Adventure, Comedies
4         Dramas, Independent Movies, Romantic Movies
5                  Documentaries, International Movies
6                                     Stand-Up Comedy
7                Anime Series, International TV Shows
8                        Dramas, International Movies
9                                       Documentaries
10  TV Action & Adventure, TV Comedies, TV Sci-Fi …
11     Horror Movies, International Movies, Thrillers
12  British TV Shows, Docuseries, International TV…
13  Children & Family Movies, Dramas, Independent …
14                                      Documentaries
15               Anime Series, International TV Shows
16   Dramas, Independent Movies, International Movies
17               Comedies, Dramas, Independent Movies
18        Dramas, Independent Movies, Romantic Movies
```

##Q7. Find Spotify songs by Olivia Rodrigo or Dua Lipa. (2 points)

[ ]:
```
c.execute('''select track_name, artist_name, popularity
from spotify_songs
where artist_name = 'Olivia Rodrigo' or artist_name = 'Dua Lipa';

''')
df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Popularity'])
```

```
display(df)
```

```
                      Track          Artist  Popularity
0            drivers license  Olivia Rodrigo          92
1                  good 4 u  Olivia Rodrigo          95
2  Levitating (feat. DaBaby)       Dua Lipa          89
3                   deja vu  Olivia Rodrigo          90
4                   traitor  Olivia Rodrigo          92
5           Don't Start Now       Dua Lipa          84
```

##Q8. Find songs on Spotify dataset that have a tempo between 100 and 120 beats per minute. (1 points)

[ ]:
```
c.execute('''
select track_name, artist_name, tempo
from spotify_songs
where tempo > 100 AND tempo < 120;
''')
df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Tempo'])
display(df)
```

```
                       Track          Artist    Tempo
0   Levitating (feat. DaBaby)       Dua Lipa  102.977
1    Kiss Me More (feat. SZA)       Doja Cat  110.970
2                      DÁKITI      Bad Bunny  109.928
3            Save Your Tears     The Weeknd  118.051
4                      Butter            BTS  109.997
5                   Dynamite            BTS  114.044
6                    traitor  Olivia Rodrigo  100.607
7          Someone You Loved   Lewis Capaldi  109.891
8                      Woman       Doja Cat  107.998
9              Qué Más Pues?       J Balvin  102.014
10                     34+35   Ariana Grande  109.978
```

##Q9. Sort a list of Spotify songs first by their popularity in descending order, and then by their danceability in ascending order. (1 points)

[ ]:
```
c.execute('''select track_name, artist_name, popularity, danceability
from spotify_songs
order by popularity desc, danceability asc;

''')
df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Popularity',␣
↪'Danceability'])
display(df)
```

```
                                      Track             Artist  \
0                                   good 4 u     Olivia Rodrigo
1                                 Bad Habits         Ed Sheeran
```

2	Woman	Doja Cat
3	Yonaguni	Bad Bunny
4	Heat Waves	Glass Animals
5	Blinding Lights	The Weeknd
6	Need to Know	Doja Cat
7	Beggin'	Måneskin
8	traitor	Olivia Rodrigo
9	drivers license	Olivia Rodrigo
10	STAY (with Justin Bieber)	The Kid LAROI
11	Sweater Weather	The Neighbourhood
12	deja vu	Olivia Rodrigo
13	Watermelon Sugar	Harry Styles
14	MONTERO (Call Me By Your Name)	Lil Nas X
15	Peaches (feat. Daniel Caesar & Giveon)	Justin Bieber
16	Save Your Tears	The Weeknd
17	INDUSTRY BABY (feat. Jack Harlow)	Lil Nas X
18	Mood (feat. iann dior)	24kGoldn
19	Levitating (feat. DaBaby)	Dua Lipa
20	Someone You Loved	Lewis Capaldi
21	you broke me first	Tate McRae
22	positions	Ariana Grande
23	I WANNA BE YOUR SLAVE	Måneskin
24	Kiss Me More (feat. SZA)	Doja Cat
25	The Business	Tiësto
26	Friday (feat. Mufasa & Hypeman) - Dopamine Re-…	Riton
27	telepatía	Kali Uchis
28	DÁKITI	Bad Bunny
29	Streets	Doja Cat
30	LA NOCHE DE ANOCHE	Bad Bunny
31	Good Days	SZA
32	Heartbreak Anniversary	Giveon
33	Leave The Door Open	Bruno Mars
34	Pareja Del Año	Sebastian Yatra
35	Head & Heart (feat. MNEK)	Joel Corry
36	Pepas	Farruko
37	WITHOUT YOU	The Kid LAROI
38	34+35	Ariana Grande
39	Don't Start Now	Dua Lipa
40	Bandido	Myke Towers
41	RAPSTAR	Polo G
42	Fiel	Los Legendarios
43	Lemonade (feat. Gunna Don Toliver & NAV)	Internet Money
44	Qué Más Pues?	J Balvin
45	Arcade	Duncan Laurence
46	Astronaut In The Ocean	Masked Wolf
47	Butter	BTS
48	Todo De Ti	Rauw Alejandro
49	Dynamite	BTS

	Popularity	Danceability
0	95	0.563
1	95	0.807
2	95	0.824
3	94	0.644
4	94	0.761
5	93	0.514
6	93	0.664
7	93	0.714
8	92	0.380
9	92	0.561
10	92	0.591
11	92	0.612
12	90	0.442
13	90	0.548
14	90	0.593
15	90	0.677
16	90	0.680
17	90	0.741
18	89	0.701
19	89	0.702
20	88	0.501
21	88	0.667
22	88	0.737
23	88	0.750
24	88	0.764
25	88	0.798
26	88	0.824
27	87	0.653
28	87	0.731
29	87	0.749
30	87	0.856
31	86	0.436
32	86	0.449
33	86	0.586
34	86	0.715
35	86	0.734
36	86	0.762
37	85	0.662
38	85	0.830
39	84	0.793
40	82	0.713
41	82	0.789
42	82	0.849
43	81	0.800
44	79	0.903
45	78	0.453

```
46          77          0.778
47          71          0.759
48          69          0.780
49          67          0.746
```

0.3 Q10. Identify Netflix shows that have a runtime longer than the average runtime of all shows on the platform. (2 points)

```
[ ]: c.execute(''' select title, duration
     from netflix_shows
     where duration > (select avg(duration)
     from netflix_shows);

     ''')
     df = pd.DataFrame(c.fetchall(), columns=['Title', 'Runtime'])
     display(df)
```

```
                                        Title  Runtime
0                                        1920      143
1                                  3 Heroines      124
2     Blue Mountain State: The Rise of Thadland       90
3                                   Blue Ruin       90
4                                  Blue Streak       94
…                                          …      …
4713                         Blood Will Tell      113
4714                                    Blow      123
4715            Blue Is the Warmest Color      180
4716                           Blue Jasmine       98
4717                               Blue Jay       81

[4718 rows x 2 columns]
```

0.4 Q11. Combine the titles of all Spotify songs and Netflix shows into a single list. Use the UNION operation to merge these two datasets, ensuring that each title appears only once in the final list. (2 points)

```
[ ]: c.execute(''' select title from netflix_shows
     UNION ALL
     select track_name from spotify_songs;

     ''')
     df = pd.DataFrame(c.fetchall(), columns=['Title'])
     display(df)
```

```
                                        Title
0                                          3%
1                                        1920
2                                  3 Heroines
```

```
3     Blue Mountain State: The Rise of Thadland
4                                 Blue Planet II
…                                            …
7832                               Qué Más Pues?
7833                   Head & Heart (feat. MNEK)
7834                                       34+35
7835                           you broke me first
7836                               Pareja Del Año

[7837 rows x 1 columns]
```

0.5 Q12. Create a list where each Spotify song is paired with each Netflix show genre. The goal is to generate combinations of songs and genres to explore how different music might fit with various types of TV shows. (1 points)

```
[ ]: c.execute(''' select track_name, artist_name, genres
     from spotify_songs, netflix_shows;

     ''')
     df = pd.DataFrame(c.fetchall(), columns=['Track', 'Artist', 'Genre'])
     display(df)
```

```
                  Track           Artist  \
0       drivers license   Olivia Rodrigo
1       drivers license   Olivia Rodrigo
2       drivers license   Olivia Rodrigo
3       drivers license   Olivia Rodrigo
4       drivers license   Olivia Rodrigo
…                   …              …
389345   Pareja Del Año  Sebastian Yatra
389346   Pareja Del Año  Sebastian Yatra
389347   Pareja Del Año  Sebastian Yatra
389348   Pareja Del Año  Sebastian Yatra
389349   Pareja Del Año  Sebastian Yatra

                                                    Genre
0       International TV Shows, TV Dramas, TV Sci-Fi &…
1          Horror Movies, International Movies, Thrillers
2             Dramas, International Movies, Sports Movies
3                                                Comedies
4       British TV Shows, Docuseries, Science & Nature TV
…                                                     …
389345                 International TV Shows, Reality TV
389346               Anime Series, International TV Shows
389347   Dramas, Independent Movies, International Movies
389348               Comedies, Dramas, Independent Movies
389349        Dramas, Independent Movies, Romantic Movies
```

```
[389350 rows x 3 columns]
```

##Q13. Determine the average popularity of Spotify songs for each genre of Netflix shows by joining the Spotify songs and Netflix shows datasets. Use an INNER JOIN to match records where genres align. (2 points)

[ ]:
```
c.execute('''select genres, avg(popularity)
from netflix_shows
INNER JOIN spotify_songs
on netflix_shows.title = spotify_songs.track_name
group by genres

''')
df = pd.DataFrame(c.fetchall(), columns=['Genre', 'Popularity'])
print(df)
```

```
Empty DataFrame
Columns: [Genre, Popularity]
Index: []
```

[ ]:
```

```